

James Philip Iddon

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Systems engineer with 8+ years' experience across the full system life cycle — requirements definition, hardware integration, verification & validation (V&V), qualification, and 24/7 mission operations — of large-scale, safety-critical sensor and instrumentation systems for international scientific programmes (50+ institutions, 6000+ members). Track record of cross-functional engineering leadership, cleanroom-based hardware integration (since 2017), and sustained mission-critical operational support. Based in Valencia.

Skills

Systems Engineering: Full system life cycle: requirements definition & traceability, architecture, integration, verification & validation (V&V), qualification, commissioning, and operational support • Configuration management and quality assurance • Cleanroom-based hardware integration • Environmental and performance qualification testing

Technical: Python • C++ • Hardware-software integration • CI/CD pipelines • Docker • Bash • Git • Linux • Automated testing and deployment • REST API • UART • I2C

Communication & Coordination: Jira • International stakeholder engagement across multi-institution programmes • Cross-functional team leadership • 24/7 operations coordination

Code Quality: Code review workflows • Static analysis tools • Testing automation and continuous integration

Professional Experience

Applied Physicist (Severo Ochoa Postdoctoral Fellow)

Apr 2026 – Present

IFIC / Universitat de Valencia, Valencia, ES

- Convener of the OBELIX CMOS sensor telescope working group for the Belle II VTX detector upgrade: coordinating systems integration across a multi-institution development team.
- Developing verification software for construction and integration of a large-scale silicon sensor subsystem (ITk Strip detector), in a cleanroom production environment.

Applied Physicist (Staff, formerly Senior Fellow)

Jul 2022 – Dec 2025

CERN, Geneva, CH

- Planned V&V for electrical characterisation of a safety-critical silicon sensor array subsystem (ATLAS ITk Pixel outer barrel, 50-institution international programme).
- Deployed and maintained 5+ automated qualification test setups for cleanroom-based silicon sensor module acceptance testing.
- Managed 24/7 operations and DevOps team for a high-reliability sensor system supporting continuous data acquisition for a large-scale international research facility (ATLAS Pixel detector, LHC).
- Provided on-call systems/DAQ engineering support for 24/7 mission-critical operations over 2 years.
- Developed and maintained data-acquisition (DAQ) software (C++/Python, CI/CD pipelines) for continuous mission operations in a safety-critical environment.
- Mentored and led 24/7 shift engineering teams drawn from a 300-person international pool.
- Chaired international engineering meetings aligning cross-functional teams across 9 time zones.
- Represented a ~30-person operations team within a 6000+ member international collaboration.
- Authored 3 novel software packages (Python), including operational tools adopted across international collaborations.
- Served as system operations coordinator for a record-setting facility operational year.

Applied Physicist (Doctorant)

Oct 2017 – Jul 2022

University of Liverpool / CERN

- Constructed, characterised, and verified 25% of the outer barrel of a large-scale silicon pixel sensor system (ALICE ITS2 — the largest monolithic pixel sensor array ever built), in ISO-standard cleanroom facilities, achieving 98% manufacturing yield.
- Developed verification software integrated with production readout and system-level V&V strategy for the full sensor system (12.5 billion pixels, 10 m² silicon).
- Validated full sensor system performance against engineering requirements.
- Performed first in-system detection-efficiency measurement using 5 million cosmic-ray tracks across the full sensor system.
- Presented technical work across a 170+ institution international collaboration, from working groups to plenary sessions, and at national and international conferences.

Education

Intensive Spanish Course

Taronja Spanish School, Valencia

Dec 2025 - May 2026

PhD in Applied Physics

University of Liverpool / CERN

Oct 2017 – Jul 2022

- Thesis: Construction, Commissioning and Performance Measurements of the Inner Tracking System Upgrade of ALICE at the LHC, focused on advanced systems engineering, integration, and comprehensive verification and validation of complex sensor systems.
- Defended June 2022. Shortlisted for the ALICE thesis award.

MPHYS Physics

University of Liverpool

Sep 2013 – Jul 2017

- Integrated masters, specialised in detector physics • Grade: First Class
- Masters project: 'Inner Tracking System Upgrade of the ALICE Experiment at the LHC', characterisation of silicon CMOS MAPS chips.

Full driving licence • Spanish residence permit • English native speaker • Spanish A2+/B1